

Technical Specification for Real-Time American Sign Language Recognition: Problem Definition and Evaluation Framework

Executive Summary

The automated recognition of American Sign Language (ASL) represents a formidable challenge at the intersection of computer vision, sequence modeling, and computational linguistics. Unlike spoken languages, which are fundamentally sequential and one-dimensional in their acoustic representation, ASL is a multi-channel, spatio-temporal language. Meaning is conveyed simultaneously through the asynchronous coordination of manual articulators—such as handshape, orientation, movement, and location—and non-manual markers, including facial expressions, head tilt, eye gaze, and mouthings¹. Consequently, mapping raw video data to semantic meaning requires a system capable of capturing high-frequency, fine-grained localized motion (such as fingerspelling) alongside low-frequency, large-scale body movements, all while operating under strict real-time computational constraints.

Historically, sign language processing systems have suffered from a modality imbalance, relying heavily on controlled studio environments and failing to generalize to the wild due to pervasive dataset leakage and a lack of true signer independence³. Modern paradigms recognize that high-fidelity sign language recognition cannot be brute-forced through general-purpose vision-language models without task-specific architectural affordances⁶.

This document serves as the foundational technical specification for defining the machine learning problem of real-time ASL recognition. It establishes a rigorous taxonomy of sign language processing tasks, differentiating between foundational capabilities and advanced semantic understanding. The specification explicitly defines the requisite input modalities—primarily prioritizing lightweight pose, hand, body, and facial landmarks, augmented by selective RGB visual features—and maps them directly to the linguistic properties of ASL. Furthermore, it outlines a robust, signer-independent evaluation framework designed to penalize dataset leakage, measure temporal segmentation accuracy, and quantify system calibration against out-of-vocabulary actions. The objective is to delineate the exact boundaries, requirements, and evaluation protocols for an initial real-time system before any specific model architectures are engineered, tested, or selected. By framing the initial problem around constrained dictionary retrieval and fingerspelling processing, this specification provides a highly viable, scalable roadmap for real-time sign language AI.

Task Taxonomy

To engineer a robust machine learning system, the problem space must be systematically

partitioned into discrete computational tasks. The following taxonomy clearly distinguishes the various sub-fields of sign language processing, detailing the theoretical nature of each task, the complexity of its underlying data distribution, and its relevance to an initial production system.

Foundational and Lexical Tasks

Static Sign Classification This task involves classifying a single, unmoving hand posture from a solitary image frame. Early literature heavily focused on this modality due to the historical limitations of compute and temporal sequence modeling. However, treating ASL as a sequence of static postures is fundamentally flawed. ASL is defined by fluid movement and trajectories; static classification is only marginally useful for a very restricted subset of the ASL alphabet or numbers. It cannot capture the dynamic nature of natural signing, nor can it account for the coarticulation that occurs between signs. Therefore, static sign classification should be viewed as a historical artifact rather than a core component of a modern pipeline.

ASL Alphabet Recognition This is the classification of the 26 letters of the English alphabet as expressed through ASL. While some letters are structurally static (e.g., "A", "B", "C"), others are intrinsically dynamic and require temporal context to distinguish (e.g., "J", "Z"). Treating alphabet recognition as a separate, isolated task is generally considered a toy problem within the computer vision community. In natural discourse, deaf individuals rarely sign isolated letters outside the context of fluid, rapid fingerspelling, making standalone alphabet recognition practically obsolete for real-world applications.

Fingerspelling Recognition Fingerspelling constitutes between 12% and 35% of natural ASL, primarily used for proper nouns, technical terminology, loan words, and concepts lacking a standardized lexical sign⁷. Fingerspelling recognition is the task of transcribing a continuous, rapid sequence of handshapes into a sequence of text characters. It requires exceptional fine-grained temporal resolution to decode heavy coarticulation, a phenomenon where the transition between letters visually blurs the canonical handshapes into a continuous motion blur. This is a critical task, distinct from whole-word recognition, often requiring sequence-to-sequence models with Connectionist Temporal Classification (CTC) or character-level decoders rather than sub-word tokenization⁹.

Isolated Sign Recognition (ISLR) ISLR is the classification of a temporally pre-segmented video clip into a single ASL gloss. A gloss is an approximate, capitalized text label used to represent the fundamental semantic unit of the sign. In this task, the system assumes the input video begins exactly at the onset of the sign and ends at its offset, eliminating the need to search for temporal boundaries. ISLR serves as the standard benchmark for evaluating spatial and temporal feature extraction¹⁰. It tests a model's ability to discriminate between thousands of vocabulary classes against diverse backgrounds and varying signers¹². High-performing ISLR systems must scale to massive vocabularies without succumbing to the long-tail class imbalance inherent in sign language lexicons¹³.

Dynamic Sign Recognition Often used interchangeably with ISLR in older literature, dynamic sign recognition explicitly differentiates itself from static classification by emphasizing motion trajectories. In modern computational paradigms, the term is largely subsumed by ISLR, as all natural isolated signs are inherently dynamic and require models like 3D-CNNs, Temporal

Convolutional Networks (TCNs), or Spatio-Temporal Graph Convolutional Networks (ST-GCNs) to process.

Continuous and Sequence-Level Tasks

Sign Spotting Sign spotting is a temporal localization and classification task. Given an unsegmented, continuous stream of signing (or a long, untrimmed video), the objective is to detect if and exactly when a specific sign or fingerspelled sequence occurs. This requires binary or multi-class classification at the frame level to establish temporal boundaries. Models typically output a sequence of Beginning-Inside-Outside (BIO) tags to demarcate the start and end of a targeted sign segment, isolating it from the surrounding movement epenthesis (the non-lexical transitional movements between signs)¹⁵.

Continuous Sign-Language Recognition (CSLR) CSLR involves translating an unsegmented, continuous video sequence of signing into a discrete sequence of corresponding glosses. Crucially, CSLR models must implicitly learn the temporal boundaries between signs and align the video frames to a discrete sequence of labels, despite severe coarticulation and varying signing speeds¹⁷. CSLR bridges the gap between isolated computer vision tasks and true linguistic sequencing, serving as an intermediate step before full translation.

Semantic and Translation Tasks

Gloss Recognition This is not a standalone computer vision task, but rather the intermediate linguistic mapping phase inherent in both ISLR and CSLR. Recognizing the gloss is distinct from translating the language. Glosses are merely meta-labels (e.g., "STORE", "GO-TO") that strip away much of the morphological richness of the sign, such as spatial agreement or adverbial intensity conveyed through facial expressions.

Sign-Language Translation (SLT) SLT is the complex task of mapping continuous sign language video directly into a grammatically correct spoken or written language (e.g., translating ASL to written English). ASL has its own distinct syntax, morphology, and grammar—such as topic-comment structure, spatial classifier predicates, and directional verbs—that do not map one-to-one with English³. SLT requires full linguistic decoding, often necessitating large language models or sequence-to-sequence transformers to bridge the vast syntactic gap between the visual-spatial modality and the linear-textual modality⁷.

Sign-Language Understanding (SLU) The highest level of abstraction, SLU encompasses tasks that require semantic and contextual reasoning over the sign language video. This includes downstream tasks such as visual question answering, semantic summarization, or sentiment analysis directly from the sign language input.

Recommended Initial Product Scope

Defining the initial product scope requires a strategic balance between user utility and current technological realities. Attempting to build a generalized Continuous Sign-Language Recognition (CSLR) or Sign-Language Translation (SLT) system as a version-one product introduces compounding points of failure. The lack of distinct temporal boundaries in continuous signing, combined with the extreme vocabulary scaling problem (where thousands of signs appear infrequently in training data), makes real-time CSLR prohibitively brittle without

a flawlessly robust foundational feature extractor.

Therefore, the initial system must aggressively constrain the problem space to ensure high accuracy, imperceptible latency, and robust signer-independent generalization. The system should initially focus on a dictionary retrieval and targeted educational assistance model, framing the machine learning problem strictly as a combination of **Isolated Dynamic Signs** and **Fingerspelling**.

1. **Isolated Sign Recognition (Vocabulary Constraints):** The system should be engineered to accept either pre-segmented video inputs or operate on a push-to-sign trigger mechanism to capture isolated glosses. To address the vocabulary scaling problem, the architecture should heavily favor one-shot or few-shot metric learning paradigms (e.g., extracting dense vector embeddings and performing rapid nearest-neighbor searches) rather than relying exclusively on fixed-class softmax layers¹⁰. This enables the system to dynamically expand its recognized vocabulary without catastrophic forgetting or full network retraining.
2. **Dedicated Fingerspelling Pathway:** Because fingerspelling relies on rapid, highly localized motion on the dominant hand, the system must utilize a specialized inference pathway for fingerspelled sequences. Fingerspelling operates at a different temporal frequency and spatial scale than standard lexical signs. By decoupling this task, the system can seamlessly switch between tracking whole-body macro-movements for standard glosses and micro-movements for alphabetical characters, greatly increasing character-level accuracy⁷.
3. **Abstention over Hallucination:** In a real-time retrieval environment, a false positive is significantly more damaging to the user experience than a non-prediction. The initial scope must heavily prioritize the detection of out-of-distribution (OOD) data. The system must reliably output an "Unknown" state when presented with non-signing gestures, motion blur, resting states, or signs that exist outside its embedded dictionary space.

Continuous signing, Sign Spotting, and full SLT are explicitly excluded from this initial scope. Attempting to parse topic-comment grammatical structures or spatial classifier predicates without first achieving >95% accuracy on isolated sign embeddings guarantees systemic failure. Once the isolated representations are highly accurate and calibrated, the system can subsequently transition to Sign Spotting as the prerequisite temporal segmentation step for handling unsegmented continuous streams.

Input and Output Specification

The extraction, normalization, and processing of visual inputs dictate the ultimate ceiling of the recognition system's accuracy. The pipeline must be designed to prioritize high-framerate, low-latency topological extraction over computationally expensive dense pixel processing wherever possible.

Required Inputs

The system will ingest an ordinary RGB video stream from a standard consumer-grade camera (e.g., a webcam or smartphone front-facing camera). From this raw visual stream, an upstream

perception module will extract specific tensor representations to feed the recognition backbones.

1. 3D Skeletal Topologies (Primary Input): The primary modality will be lightweight topological keypoints extracted via advanced pose estimation pipelines. Processing coordinate data requires exponentially fewer floating-point operations than processing raw video volumes, enabling the system to meet its real-time constraints¹⁹.

- *Hand Kinematics:* Extraction of both the dominant and non-dominant hand keypoints (typically 21 spatial coordinates per hand, plotted in 3D space to capture depth and self-occlusion).
- *Upper Body Pose:* Tracking of the shoulders, elbows, wrists, and torso center. This is mandatory for establishing the absolute scale of the user and the trajectory of the movement through the signing space.
- *Dense Facial Landmarks:* A subset of facial nodes specifically isolating the eyebrows, eyelids, and outer/inner lip contours.
- *Spatial Normalization:* These raw coordinates must undergo rigorous spatial normalization relative to a stable physiological anchor (e.g., standardizing the distance between the shoulders to a value of 1, and centering the coordinate system on the torso). This ensures the downstream network is invariant to the camera distance, user height, and arbitrary subject positioning within the frame²⁰.

2. RGB Visual Cropping (Secondary/Fallback Input): While skeletal data is highly efficient, pose estimators frequently fail during rapid motion blur or severe self-occlusion (e.g., when the hand crosses directly over the face). Furthermore, pixel-based features contain crucial texture and micro-tension information that abstract coordinates lack.

- *Dynamic Hand Crops:* The system will utilize dynamic bounding-box crops of the dominant hand. These crops will be triggered selectively during known fingerspelling sequences, or dynamically when the upstream pose-estimation confidence metrics fall below a predefined safety threshold²¹.
- *Temporal Resolution:* The temporal resolution for all inputs must be maintained at a strict minimum of 30 frames per second (FPS). Any frame dropping or temporal downsampling below this threshold introduces temporal aliasing, which effectively destroys the transitional frames necessary to decode rapid coarticulation in fingerspelling tasks²².

Output Specifications

Because the system manages disparate linguistic tasks, the network architecture must bifurcate at the prediction head, projecting the shared spatio-temporal features into task-specific output spaces.

1. Isolated Sign Output Space:

- The primary output for an isolated sign is a high-dimensional dense embedding vector. This vector mathematically represents the spatio-temporal essence of the input sequence. The vector is subsequently compared against a pre-computed vector database of known gloss embeddings using a distance metric (such as Cosine Similarity or Euclidean distance). The output is the Top- K closest matching glosses¹⁰.

- A fully calibrated confidence score (ranging between 0.0 and 1.0) must accompany the prediction, indicating the exact statistical likelihood that the generated embedding genuinely matches the retrieved known sign.

2. Fingerspelling Output Space:

- The output is a continuous character-level sequence string. Because fingerspelling requires the model to reason about the co-occurrences of letters without explicit temporal boundaries between individual characters, the output head cannot use standard softmax classification. It must utilize Connectionist Temporal Classification (CTC) over a character vocabulary, or an autoregressive token-free mechanism (e.g., character-level ByT5 architectures) to handle the fluid, unsegmented spelling⁷.

3. Boundary/Spotting Output Space (For Future Phasing):

- While spotting is a secondary phase, the architecture must support frame-level probabilistic outputs utilizing a BIO (Beginning, Inside, Outside) tagging scheme. This outputs a temporal map demarcating the exact start and end frames of a sign sequence¹⁵.

Task Unification and Shared Representations

These distinct tasks do not require entirely separate end-to-end pipelines. They can, and should, share a foundational spatio-temporal encoder. A hybrid architecture—such as a ResNet-Transformer combination, or a Spatio-Temporal Graph Convolutional Network (ST-GCN) processing the skeletal sequence—can compute a rich, unified latent representation of the video¹¹. The shared backbone learns the fundamental physics and phonology of ASL motion. The divergence occurs only at the final output heads: an embedding projector for ISLR, a sequence decoder for fingerspelling, and a frame-wise classifier for temporal boundary detection.

ASL Linguistic Requirements

A pervasive fallacy in computer vision approaches to sign language processing is treating the hands as the exclusive source of semantic information. American Sign Language is a deeply compositional system where meaning arises from the simultaneous execution of multiple sublexical phonological parameters. If a recognition system is blind to any of these parameters, it will fundamentally fail to distinguish between vast arrays of minimal pairs (distinct signs that differ in only one phonological parameter)¹.

The Five Phonological Parameters

Based on established linguistic models (such as Stokoe's original classification and the expanded parameters of Battison, Liddell, and Johnson), the system must be capable of capturing the following parameters to ensure semantic fidelity²:

1. **Handshape:** The specific configuration of the fingers and thumb (e.g., open palm, closed fist, curved fingers, index isolation). ASL utilizes approximately 40 to 50 distinct handshapes. Accurately capturing these requires high-fidelity tracking of all 21 joints per hand, as well as the spread and flexion of individual digits¹⁴.
2. **Location (Place of Articulation):** The precise spatial placement of the sign relative to

the body (e.g., the forehead, chest, chin, or the non-dominant hand). Distinguishing between the sign for "FATHER" (thumb on the forehead) and "MOTHER" (thumb on the chin) requires robust spatial grounding against the face and torso².

3. **Movement:** The macro path trajectory and the internal joint movement dynamics (e.g., wrist twisting, repeated oscillatory motion, arc paths). Tracking this requires precise temporal monitoring of the shoulders, elbows, and wrists to derive velocity and acceleration vectors²⁵.
4. **Palm Orientation:** The direction in which the palm and fingertips face. This relies heavily on capturing the radioulnar articulation (wrist rotation) and the 3D depth geometry of the hand relative to the camera²⁵.
5. **Non-Manual Markers (NMMs):** The most frequently ignored yet linguistically critical parameter. NMMs dictate grammar, syntax, and adjective severity. Without them, a system cannot distinguish between a statement and a question.

Requisite Modalities for Linguistic Mapping

To adequately represent the aforementioned linguistics, the system must extract distinct features from various anatomical regions. Relying solely on hand tracking will result in catastrophic information loss.

- **Hands (Fingers, Palms, Wrists):** Absolutely essential for determining handshape and orientation.
- **Arms (Elbows, Shoulders):** Essential for tracking the kinematics of the movement path and providing scale context. The elbow angle often dictates the severity or size of a signed concept.
- **Torso (Shoulders to Hips):** Serves as the spatial reference frame, creating the bounded "signing space." Without the torso reference, absolute hand coordinates floating in space are mathematically meaningless.
- **Face and Head (Eyebrows, Eyes, Mouth, Head Tilt):** Strictly required for Non-Manual Markers.
 - *Eyebrows & Head:* Raised eyebrows accompanied by a forward head tilt indicate yes/no questions; furrowed eyebrows indicate "wh-" questions (who, what, where). Head tilt and shoulder shifting are also used to indicate spatial relationships, conditional clauses, or a change in narrative perspective (role-shifting).
 - *Eye Gaze:* Used for spatial indexing. A signer will establish a noun in a physical space next to them, and subsequent eye gazes toward that empty space function grammatically as pronouns.
 - *Mouth:* ASL uses specific native mouth morphemes (e.g., "cha" for something abnormally large, "th" for something careless or inattentive) that completely alter the meaning of the manual sign. Furthermore, partial English mouthings are sometimes used alongside signs for lexical disambiguation of homophones.

Architectural Trade-offs in Phonological Perception

Recent studies on the phonological perception of deep learning models reveal clear architectural trade-offs. While skeletal pose estimation effectively captures the intricate joint

articulations required for handshape contrasts, pose estimators often discard the localized visual textures and micro-spatial interactions that dictate precise locations. Pixel-based (RGB) models demonstrate superior sensitivity to location changes (e.g., determining if a hand is resting exactly flush against the cheek versus hovering a centimeter in front of it) and fine-grained facial muscle tension². Therefore, the system must employ a hybrid modality strategy: passing the full skeletal topology as the primary, low-latency input, while selectively injecting RGB hand crops or facial region-of-interest (ROI) features to disambiguate tight minimal pairs or verify contact points¹¹.

Real-Time System Requirements

To ensure the system functions adequately in a real-time conversational setting, or provides immediate feedback in an interactive educational application, extremely stringent computational constraints must be established. Algorithms that achieve state-of-the-art accuracy but require seconds to process a video clip are unusable in production.

- **End-to-End Latency:** "Real-time" in this context dictates that the system must return a finalized prediction within 150 to 200 milliseconds after the physical completion of the sign (the offset boundary). A delay exceeding 300 milliseconds disrupts conversational flow, leading to user frustration and communication breakdown. This latency budget includes frame capture, pose extraction, network inference, and output rendering.
- **Throughput:** The system must process inputs at no less than 30 frames per second. Processing at lower frame rates inherently drops critical transitional frames. Because fingerspelling involves micro-movements occurring in fractions of a second, downsampling the frame rate leads to catastrophic failure in character recognition²².
- **Memory Footprint:** Assuming edge-deployment on consumer mobile hardware, tablets, or standard web browsers (to ensure privacy and offline capability), the total model size must be strictly bounded. The entire pipeline—including upstream pose estimation and downstream recognition—must not exceed 2 GB of RAM. The backbone architecture should leverage parameter-efficient designs, targeting under 50 million parameters. Recent advancements in Spatio-Temporal Graph Convolutional Networks (like SignBart) have demonstrated that exceptional accuracy can be achieved with sub-10 million parameters by independently encoding X and Y coordinates with cross-attention¹⁹.
- **Temporal Windowing Mechanics:** The system must utilize sliding window buffers or streaming stateful architectures (such as stateful RNNs or streaming Transformers) to avoid recalculating features for overlapping temporal segments. This ensures that the computational complexity remains $O(1)$ per new incoming frame, preserving the latency budget as the video stream lengthens indefinitely.

Evaluation Framework

Evaluation protocols in sign language research have historically been fundamentally flawed, leading to severe overestimations of model capability and a complete failure of models to generalize to real-world deployment. The following framework mandates strict protocols and specialized mathematical metrics to ensure absolute rigor.

Combating Dataset Leakage and Signer Overlap

Dataset leakage is the most pervasive technical error in sign language machine learning. Due to the limited size of early datasets, many studies utilized random train/test splits, resulting in the same signer appearing in both the training data and the test data. Deep learning models, optimizing for the path of least resistance, frequently learn to classify the identity of the signer, the specific color of their clothing, the background room geometry, or their resting facial expression, rather than the intrinsic linguistics of the sign itself³.

When signer overlap is present in evaluation sets, binary sign detection accuracy can artificially inflate by over 6%, and complex classification tasks inflate significantly more⁴. **All evaluation must be conducted using strictly Signer-Independent splits.** A model trained on Signers A through Y must only be tested on Signer Z, who has never been observed by the network¹².

Furthermore, for continuous tasks, models must be tested on Unseen Sentences (combinations of glosses never seen together during training) to prevent the network from overfitting to specific, highly probable lexical sequences (e.g., memorizing the phrase "HOW ARE YOU" as a single monolithic block)²⁷.

Evaluation Metrics

The system will utilize the following array of metrics, chosen specifically to handle the diverse, multifaceted nature of ASL tasks.

Metric	Task Application	Mathematical and Conceptual Justification
Top-1 and Top-5 Accuracy	ISLR	Measures the frequency with which the correct gloss is the highest predicted probability (Top-1) or within the top 5 predictions. Top-5 is crucial for dictionary retrieval UX, where users are presented with a shortlist of intended signs to select from ²⁸ .
Macro F_1 Score	ISLR / Classification	Given the extreme long-tail distribution of ASL vocabularies (severe class imbalance), raw accuracy is mathematically deceptive. A model can achieve high accuracy by only learning

		the 50 most common signs. Macro F_1 computes the unweighted mean of the F_1 scores for each individual class, ensuring that rare signs are weighted equally against highly frequent signs.
Per-class F_1 Score	ISLR / Classification	Calculates the harmonic mean of precision and recall for each specific gloss. This is essential for diagnosing systemic failures in specific phonological clusters (e.g., discovering the model fails entirely on signs utilizing the 'V' handshape).
Semantic F_1 Score	ISLR / SLU	Conventional F_1 treats semantically related misclassifications (e.g., predicting the sign "CAR" instead of "DRIVE") as total failures, functionally identical to predicting a completely unrelated sign like "APPLE". Semantic F_1 utilizes a predefined label similarity matrix to grant partial mathematical credit for semantically related predictions, better reflecting fuzzy category boundaries, annotator disagreement, and real-world conversational

		utility ²⁹ .
Character Error Rate (CER)	Fingerspelling	Computed as $CER = \frac{S + D + I}{N}$, where S, D, and I represent character substitutions, deletions, and insertions via Levenshtein distance, and N is the total number of ground truth characters. CER is strictly superior to Word Error Rate for fingerspelling; missing a single rapid letter in a fingerspelled technical noun still provides immense phonetic value, which CER captures but WER penalizes completely⁷.
Word Error Rate (WER) / Sign Error Rate	CSLR / SLT	The standard metric for continuous recognition, utilizing the same Levenshtein distance formulation as CER, but applied at the discrete gloss or word level.
Boundary F_1 (BF) / Segmental F_1	Sign Spotting / Segmentation	Evaluates temporal localization accuracy. A predicted sign segment is considered correct if its temporal Intersection-over-Union (IoU) with the ground truth exceeds a predefined threshold. Boundary F_1 specifically measures how closely the predicted

		boundary frame matches the ground truth boundary, accommodating minor human annotation variances in marking exact start/stop frames ³⁰ .
Mean Average Precision (mAP)	Sign Spotting	Evaluates the area under the precision-recall curve for localized sign predictions across multiple IoU thresholds.
Expected Calibration Error (ECE)	Real-Time Confidence	Measures the difference between the model's output confidence and its actual empirical accuracy. A well-calibrated model outputting 80% confidence should be correct exactly 80% of the time. ECE is vital for a real-time system to mathematically determine when to abstain from presenting an uncertain prediction to the user ³³ .
Unknown-class AUROC	Open Set Recognition	The Area Under the Receiver Operating Characteristic curve for detecting Out-of-Distribution (OOD) data. The system must reliably discriminate between the 2,000+ known signs and the infinite, unbounded space of unknown signs or random, non-linguistic human movements.

Dataset Requirements

To train a resilient, highly generalized model, the datasets utilized (such as MS-ASL, WLASL, ChicagoFSWild, and specifically ASL Citizen) must exhibit vast diversity and scale¹². The data curation pipeline must strictly enforce the following characteristics:

Characteristic	Requirement Description
Number of Signers	A strict minimum of 50 to 100 distinct signers is required to prevent the model from memorizing specific body morphologies, idiosyncrasies, and personal signing speeds ¹² .
Signer-Independent Splits	Absolute, verifiable segregation of signers across training, validation, and test sets to guarantee zero data leakage ¹² .
Camera & Background Variation	Data must be elicited via crowdsourcing to ensure models learn to filter out visual noise, complex backgrounds, and varying aspect ratios (e.g., standard webcams vs. vertical smartphones) rather than relying on the pristine purity of green-screen studio environments ¹² .
Lighting & Skin-Tone Diversity	Crucial for the robustness of upstream pose estimators. If a dataset biases heavily toward light skin tones, the hand-tracking algorithms will catastrophically fail for users with darker skin, particularly in low-contrast lighting scenarios.
Signing Speed	Must capture naturalistic speed variances. Fluent, native signers use heavily compressed motion paths and high-velocity transitions compared to the exaggerated, slow motions of ASL learners.
Handedness	Must explicitly include left-hand dominant signers. Models must learn to generalize to

	lateral inversions of the signing space without failing.
Occlusion	Datasets must contain natural self-occlusion (hands crossing the face or torso) to force the network to rely on temporal context rather than singular pristine frames.
Frame Rate & Resolution	Minimum 30 FPS to prevent temporal blurring of fingerspelling. Minimum 720p resolution to ensure sufficient pixel density for extracting minute facial expressions and dense finger configurations.
Annotation Quality	Temporal boundaries must be meticulously labeled by fluent annotators. Sublexical parameters (handshape, location) should preferably be cross-referenced and aligned with established linguistic corpora (e.g., ASL-LEX 2.0) to enable deep phonological analysis ¹ .

Major Technical Risks

1. **Coarticulation and Movement Epenthesis:** In natural, fluid signing, the hands do not return to a resting state between signs. The transitional movement from the end of Sign A directly to the beginning of Sign B creates an artificial, non-lexical trajectory known as movement epenthesis. A major technical risk is that the neural network will erroneously learn these transitions as fundamental components of the signs themselves, causing complete failure when the sign is preceded by a different vocabulary word¹⁵.
2. **Upstream Pose Estimation Failures:** Relying heavily on 3D skeleton tracking introduces a critical upstream dependency. Motion blur during high-velocity signs, or severe self-occlusion, frequently causes current consumer-grade pose estimators to drop frames or wildly hallucinate joint positions. If the input coordinates are corrupted, the downstream recognition network receives adversarial noise.
3. **The Vocabulary Scaling Bottleneck:** ASL contains tens of thousands of signs, many of which only appear a handful of times in available datasets. Standard softmax classification fails to scale to this long-tail distribution, necessitating highly complex few-shot metric learning architectures that are difficult to tune for real-time edge deployment¹³.
4. **Semantic Ambiguity of Sublexical Parameters:** Similar handshapes can mean entirely

different things based on subtle spatial positioning (location minimal pairs). If the skeletal normalization algorithms aggressively alter absolute scale to make users uniform, this delicate spatial nuance may be mathematically scrubbed from the data, destroying the model's ability to differentiate the signs.

Recommended Research Boundaries

During the initial development and prototyping phase, the following research avenues should be strictly bounded, avoided, or deferred to ensure engineering resources are not squandered on presently intractable problems:

- **End-to-End Large Language Model Translation:** Do not attempt to map raw video directly to LLM text tokens for full English translation. Current zero-shot Vision-Language Models (VLMs) are highly inaccurate for ISLR, let alone continuous translation, as they lack the spatial inductive biases necessary for ASL phonology⁶. The focus must remain exclusively on extracting robust spatio-temporal representations.
- **Avatar Generation and Sign Language Production:** Generative models for synthesizing sign language avatars (e.g., predicting SMPL-X motion parameters from text) introduce entirely different complexities regarding naturalism, inverse kinematics, and the "uncanny valley" effect¹. This is entirely out of scope for a recognition system.
- **Grammar and Syntax Mapping:** Attempting to hard-code ASL linguistic rules (like spatial verb agreement or classifier predicates) into the neural network architecture will fail. The system must learn these representations dynamically from the data in later continuous phases, rather than through rigid rule-based engineering.

Handoff to the Next Research Stage

This document formally defines the bounds of the real-time ASL recognition problem. For the subsequent research phase (Model Architecture Selection and Prototype Training), the engineering and machine learning teams should proceed under the following strict directives:

Final Task Definitions: The system is fundamentally an information retrieval and rapid transcription pipeline operating on spatio-temporal video data, executing two primary operations: mapping isolated motion sequences to semantic gloss embeddings, and transcribing highly localized, rapid hand motion into character sequences.

Recommended Initial Tasks:

1. Signer-Independent Isolated Sign Recognition (ISLR) framed as dense vector search and dictionary retrieval.
2. Continuous Fingerspelling Recognition utilizing token-free, character-level sequence decoding.

Required Input Modalities:

1. Normalized 3D Skeletal Data: 21-point hand meshes, upper body pose (arms, shoulders, torso), and dense facial landmarks (eyebrows, mouth, eyes).
2. Dynamic RGB Cropping: Fast, localized pixel extraction conditioned on pose-confidence thresholds or fingerspelling triggers.

Required Output Labels:

1. A high-dimensional spatial-temporal embedding vector for the sign sequence.
2. A sequence of characters (for fingerspelling).
3. A calibrated confidence score (0.0 - 1.0) dictating abstention.

Evaluation Metrics:

- Top-1 / Top-5 Accuracy
- Macro F_1 Score and Per-class F_1
- Character Error Rate (CER)
- Expected Calibration Error (ECE)
- Unknown-class AUROC

Real-Time Constraints:

- End-to-end latency: ≤ 200 ms.
- Throughput: ≥ 30 FPS.
- Total System Memory: ≤ 2 GB total footprint.

Open Questions Specifically Related to Pose and Skeleton Representation:

1. How should missing or hallucinated joints caused by self-occlusion in upstream pose estimators be mathematically imputed, smoothed, or masked within the temporal sequence?
2. What is the optimal spatial normalization strategy to render the skeleton invariant to camera distance without destroying the relative spatial location coordinates necessary for resolving location-based minimal pairs?
3. Can highly efficient Graph Convolutional Networks (GCNs) effectively model the asynchronous interactions between the facial nodes and hand nodes, or is a full Attention/Transformer mechanism required across the skeletal graph?

List of the 15 Most Important Sources for the Next Stage:

1. *ASL Citizen: A Community-Sourced Dataset for Advancing Isolated Sign Language Recognition* (Desai et al.)¹².
2. *Sign-Language Datasets at Scale: A Comprehensive Survey on Resources* (Dataset fragmentation and scaling)⁵.
3. *Investigating Signer-Independent Sign Language Recognition on the LSA64 Dataset* (Signer-dependence accuracy drops)³.
4. *Effect of Signer Overlap on Sign Language Detection* (Dataset Leakage quantification)⁴.
5. *SignBart: New approach with the skeleton sequence for Isolated Sign language Recognition* (X/Y coordinate separation and efficiency)¹⁹.
6. *MS-ASL: A Large-Scale Data Set and Benchmark for Understanding ASL*³⁴.
7. *FluentSigners-50: A Large-Scale Signer Independent Dataset* (Real-life variability and continuous signing)²⁷.
8. *American Sign Language fingerspelling recognition in the wild* (ChicagoFSWild dataset introduction)²³.
9. *HandReader: Advanced Techniques for Efficient Fingerspelling Recognition* (TSAM and

RGB+Keypoint fusion)²¹.

10. *Evaluating character-level vs subword-level tokenization for fingerspelling* (ByT5 vs T5 advantages)⁷.
11. *Sign Language Segmentation with Temporal Convolutional Networks* (Boundary/Segmental F1 and IOUs)³⁰.
12. *Semantic F1 Scores: Evaluation metrics for subjective or fuzzy multi-label classification* (Partial credit for semantics)²⁹.
13. *One-Shot Learning for Isolated Sign Language Recognition across Languages* (Dense vector searching for vocabulary scaling)¹⁰.
14. *Sign language phonological parameters and minimal pairs* (ASL-LEX 2.0) (Handshape, location, movement structures)¹.
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